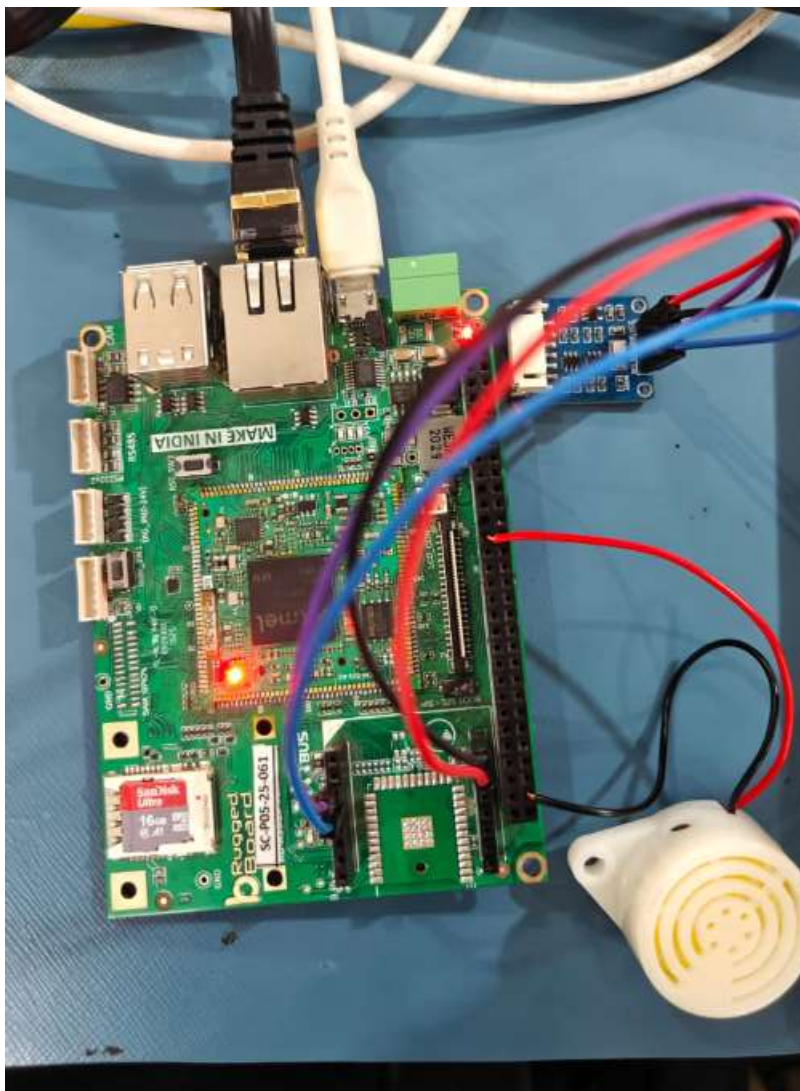


Factory Management System- using RB a5d2x and Thingsboard IOT platform

This project uses a BME680 sensor connected to a Rugged Board running Embedded Linux. Sensor data is transmitted to ThingsBoard using HTTP telemetry. A local SFM-27 buzzer alarm is activated when temperature exceeds 30°C.

Pictures:



BME680 connected to Rugged Board via I2C on mikroBUS.

SFM-27 buzzer connected to RuggedBoard via GPIO pins.

```

root@rugged-board-a5d2x:/run# i2cdetect -y 0
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
20:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
40:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
50: UU 51 52 53 54 55 56 57 -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- -- -- UU --
70: -- -- -- -- -- -- -- 77

```

BME680 is detected at address 0x77.

Sensor Output:

```

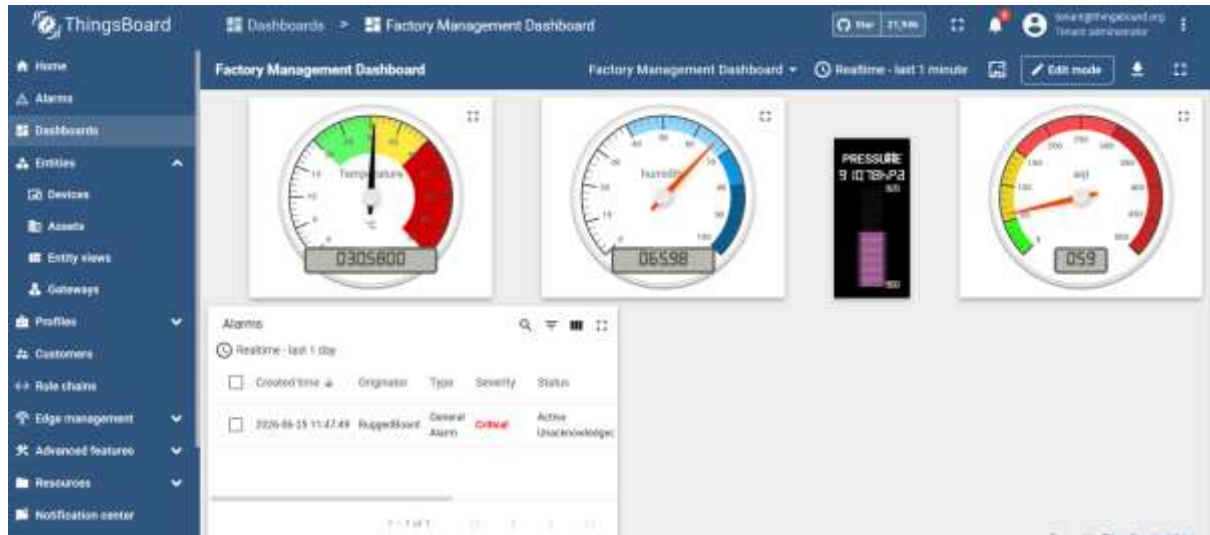
=====
Temp : 27.99 C
Hum  : 62.03 %
Pres : 910.67 hPa
Gas  : 12496 ohm
AQI  : 50
=====

```

Telemetry updated from the RuggedBoard to Thingsboard:

RuggedBoard			
Device details			
Details	Attributes	Latest telemetry	Calculated fields
Alarms	Events		
Telemetry			
☐	Last update time	Key ↑	Value
☐	2026-06-25 11:50:21	alarm	0
☐	2026-06-25 11:50:21	aqi	54
☐	2026-06-25 11:50:21	humidity	61.76
☐	2026-06-25 11:50:21	pressure	910.69
☐	2026-06-25 11:50:21	temperature	28.11

Factory Management Dashboard on Thingsboard:



Alarm system in thingsboard:

Alarms

Realtime - last 1 day

☐	Created time	Originator	Type	Severity	Status	Assigned
☐	2025-06-25 11:47:49	RuggedBoard	General Alarm	Critical	Active	Unassigned